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A2**Started on** Friday, 2 October 2026, 7:46 PM**State** Finished**Completed on** Friday, 2 October 2026, 7:47 PM**Time taken** 58 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	g	t	a	b
s2		g	x	t	x	a	y b

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char a[100], b[100];
6      int dp[101][101];
7
8      scanf("%s", a);
9      scanf("%s", b);
10
11     int n = strlen(a);
12     int m = strlen(b);
13
14     for (int i = 0; i <= n; i++)
15         dp[i][0] = 0;
16
17     for (int j = 0; j <= m; j++)
18         dp[0][j] = 0;
19
20     for (int i = 1; i <= n; i++) {
21         for (int j = 1; j <= m; j++) {
22             if (a[i - 1] == b[j - 1])
23                 dp[i][j] = dp[i - 1][j - 1] + 1;
24             else
25                 dp[i][j] = dp[i - 1][j] > dp[i][j - 1] ?
26                     dp[i - 1][j] : dp[i][j - 1];
27         }
28     }
29
30     printf("%d", dp[n][m]);
31
32     return 0;
33 }
```

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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